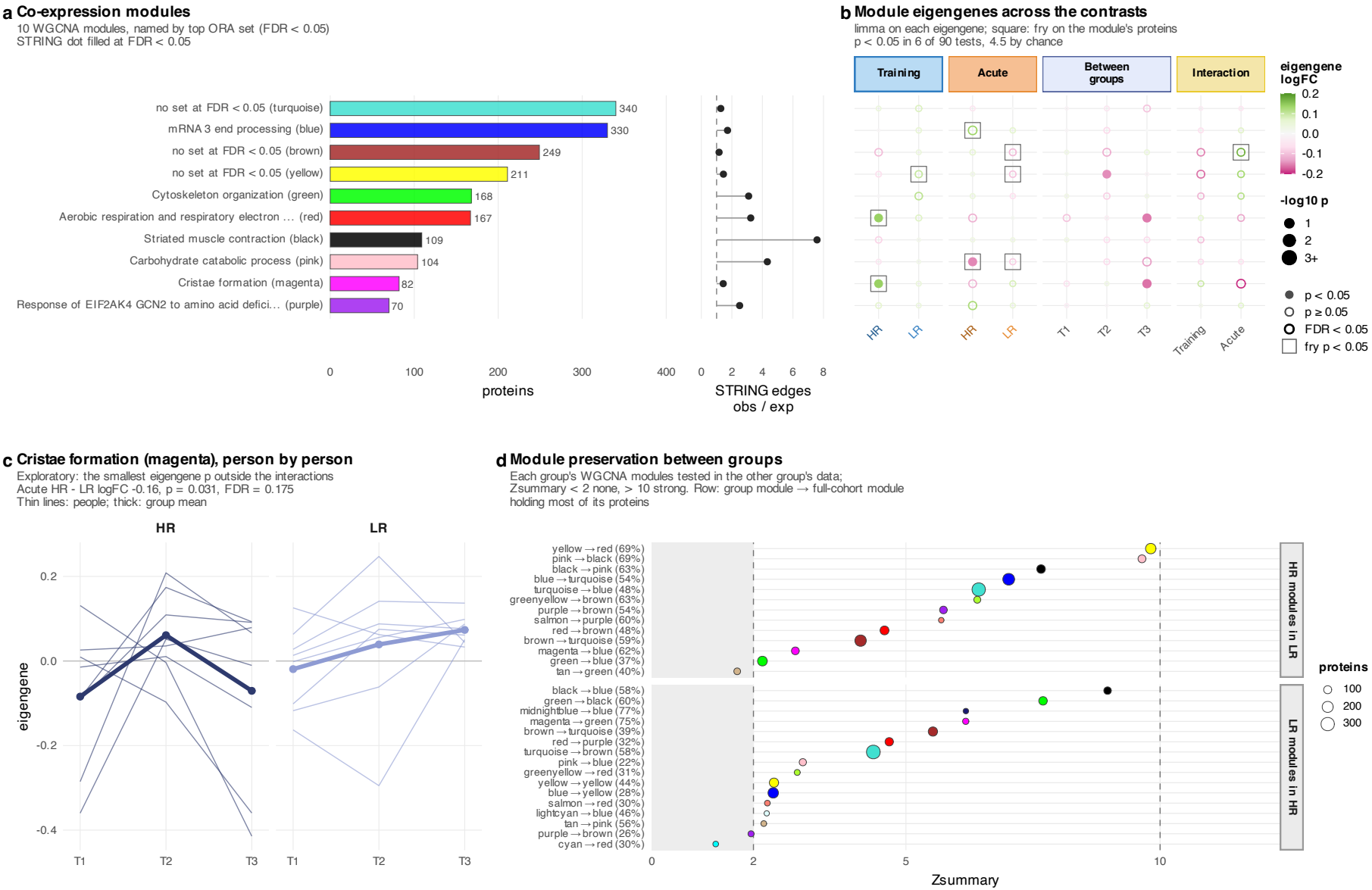
a Co-expression modules
10 WGCNA modules, named by top ORA set (FDR < 0.05)
STRING dot filled at FDR < 0.05
no set at FDR < 0.05 (turquoise)
mRNA 3 end processing (blue)
no set at FDR < 0.05 (brown)
no set at FDR < 0.05 (yellow)
Cytoskeleton organization (green)
Aerobic respiration and respiratory electron … (red)
Striated muscle contraction (black)
Carbohydrate catabolic process (pink)
Cristae formation (magenta)
Response of EIF2AK4 GCN2 to amino acid defici… (purple)
340
330
249
211
168
167
109
104
82
70
0
100
200
300
400
proteins
0
2
4
6
8
STRING edges
obs / exp
b Module eigengenes across the contrasts
limma on each eigengene; square: fry on the module's proteins
p < 0.05 in 6 of 90 tests, 4.5 by chance
Training
Acute
Between groups
Interaction
HR
LR
HR
LR
T1
T2
T3
Training
Acute
eigengene logFC
0.2
0.1
0.0
-0.1
-0.2
-log10 p
1
2
3+
p < 0.05
p ≥ 0.05
FDR < 0.05
fry p < 0.05
c Cristae formation (magenta), person by person
Exploratory: the smallest eigengene p outside the interactions
Acute HR - LR logFC -0.16, p = 0.031, FDR = 0.175
Thin lines: people; thick: group mean
HR
LR
eigengene
0.2
0.0
-0.2
-0.4
T1
T2
T3
T1
T2
T3
d Module preservation between groups
Each group's WGCNA modules tested in the other group's data;
Zsummary < 2 none, > 10 strong. Row: group module → full-cohort module
holding most of its proteins
yellow → red (69%)
pink → black (69%)
black → pink (63%)
blue → turquoise (54%)
turquoise → blue (48%)
greenyellow → brown (63%)
purple → brown (54%)
salmon → purple (60%)
red → brown (48%)
brown → turquoise (59%)
magenta → blue (62%)
green → blue (37%)
tan → green (40%)
black → blue (58%)
green → black (60%)
midnightblue → blue (77%)
magenta → green (75%)
brown → turquoise (39%)
red → purple (32%)
turquoise → brown (58%)
pink → blue (22%)
greenyellow → red (31%)
yellow → yellow (44%)
blue → yellow (28%)
salmon → red (30%)
lightcyan → blue (46%)
tan → pink (56%)
purple → brown (26%)
cyan → red (30%)
HR modules in LR
LR modules in HR
0
2
5
10
Zsummary
proteins
100
200
300